SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP

C. SMTP
D. SNMP

Questions:

1. Select the correct HTML tag for video embedding and justify your answer.
2. Identify the appropriate HTTP method for form submission and explain why.
3. Choose the proper semantic element for navigation and justify its usage.
4. Identify the correct XML syntax and explain the error in other options.
5. Choose the correct protocol for web communication and justify your choice.

. Correct HTML tag for video embedding

Correct Answer: Option B

<video src="video.mp4" controls></video>

Justification: The <video> tag is used to embed videos in an HTML webpage. The controls attribute displays video controls such as play, pause, and volume.

2. Appropriate HTTP method for form submission

Correct Answer: Option B — POST

<form action="/register" method="POST">

<input type="text" name="username">

<input type="password" name="password">

<button type="submit">Register</button>

</form>

Justification: The POST method sends form data to the server in the request body. It is more appropriate for submitting registration data than GET because the data is not appended to the URL. However, HTTPS is required to encrypt the data during transmission.

3. Proper semantic element for navigation

Correct Answer: Option B

<nav>

Home

About

```
    <a href="contact.html">Contact</a>
</nav>
```

Justification: The <nav> element represents a section containing major navigation links. It improves the semantic structure and accessibility of a webpage.

4. Correct XML syntax

Correct Answer: Option B

```
<note>
    <to>User</to>
    <from>Admin</from>
</note>
```

Justification: XML requires properly nested elements, matching opening and closing tags, and a single root element.

Errors in the other options:

- Option A: The root element <note> is not closed.
- Option C: The <to> opening tag is incorrectly closed with </from>, and the XML elements are mismatched.
- Option D: The <note> element is not properly closed.

5. Correct web communication protocol

Correct Answer: Option B — HTTP

Justification: HTTP (Hypertext Transfer Protocol) is used for communication between web browsers and web servers. HTTPS is the secure version of HTTP and should be used for modern web applications that handle sensitive user data.

Code of Conduct:

I G.Akash 192372239 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student